

Project Initialization and Planning Phase

Date: 15 March 2024

Team ID: (Enter your Team ID)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

Team Members

S.No	Name	Role	Responsibility
1	G Naveen	Team Lead	Project planning, model development, GitHub management
2	Chandana Busireddy	Member	Dataset collection, preprocessing and documentation
3	Gandham Joshna	Member	Data visualization and Exploratory Data Analysis
4	Nandineni Gowthami	Member	Flask web application and testing
5	Shaik Talha	Member	Model training, report preparation and project presentation

Customer Problem Statement

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Government Policy Maker	Evaluate the development level of different countries	Manual analysis of multiple indicators is time-consuming	HDI depends on health, education, and income together	Confident that an AI-based prediction system will improve decision-making

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-2	A Research Analyst	Compare HDI values across countries	Large datasets are difficult to analyze manually	Multiple socio-economic factors influence development	Better informed through accurate predictions
PS-3	A Student	Understand how HDI is calculated	Different development indicators are difficult to interpret	The relationship between indicators is complex	More confident in learning global development concepts

Project Description

The Human Development Index (HDI) Prediction System is a Machine Learning application developed to predict the human development category of a country based on four important socio-economic indicators:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) Per Capita

The system classifies countries into one of four categories:

- Very High HDI
- High HDI
- Medium HDI
- Low HDI

The prediction model helps governments, researchers, policymakers, and educational institutions analyze development levels efficiently.

Team Brainstorming

Team Member	Suggestion
G Naveen	Develop a Machine Learning model for HDI prediction.
Chandana Busireddy	Collect and preprocess HDI datasets from reliable sources.
Gandham Joshna	Create graphs and visualizations for better understanding.
Nandineni Gowthami	Design a Flask-based web application.
Shaik Talha	Test the model and prepare project documentation and presentation.

Final Selected Idea

Human Development Index (HDI) Prediction Using Machine Learning

The team selected this idea because it combines Machine Learning with socio-economic data to provide accurate HDI predictions, helping users understand development levels and supporting evidence-based decision-making.

Expected Benefits

- Predicts HDI accurately.
- Saves time compared to manual analysis.
- Supports government policy decisions.
- Helps researchers compare countries.
- Provides an easy-to-use web application.
- Improves understanding of human development indicators.

Team Contribution Summary

Name	Contribution
G Naveen	Team coordination, GitHub repository management, model integration

Name	Contribution
Chandana Busireddy	Dataset collection and preprocessing
Gandham Joshna	Data visualization and analysis
Nandineni Gowthami	Flask application development and testing
Shaik Talha	Machine learning model training, documentation, GitHub updates, and final presentation